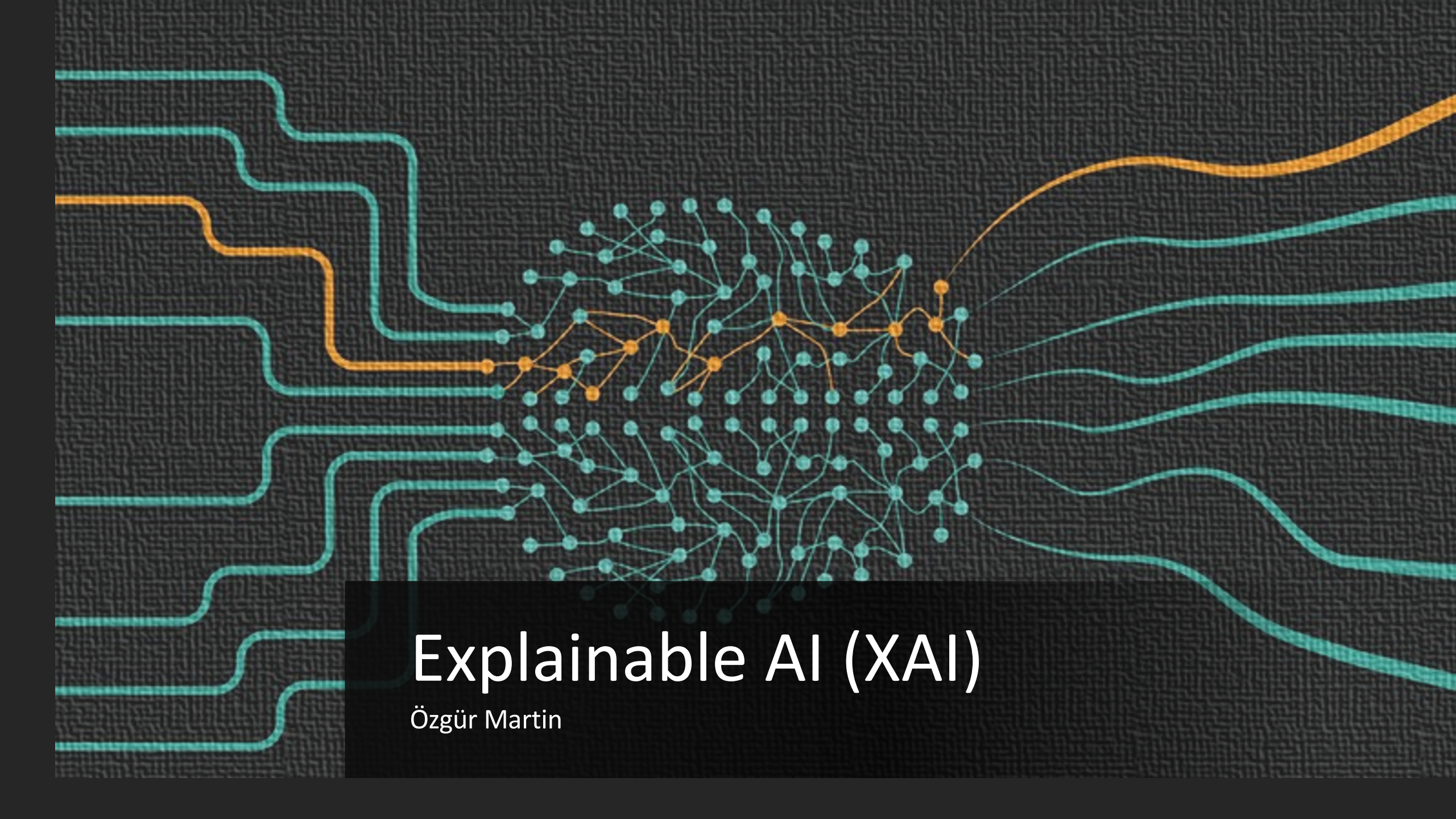


The background of the slide features a dark, textured surface. On the left, several teal lines enter from the edge and branch out in a circuit-like fashion. In the center, there is a dense network of small teal nodes connected by thin lines, with a few orange nodes and lines weaving through it. On the right, a single orange line curves upwards, while several teal lines flow outwards in a wavy pattern.

Explainable AI (XAI)

Özgür Martin

The Washington Post

Democracy Dies in Darkness

‘Creative ... motivating’ and fired



Sarah Wysocki was out of work for only a few days after she was fired by DCPS last year. She is now teaching at Hybla Valley Elementary School in Fairfax County. (Jahi Chikwendiu/The Washington Post)

By **Bill Turque**
March 6, 2012

HDSR

Why Are We Using Black Box Models in AI When We Don't Need To? A Lesson From An Explainable AI Competition

by Cynthia Rudin and Joanna Radin

Published on Nov 22, 2019

 **CGAP**

BLOG 05 SEPTEMBER 2019

Algorithm Bias in Credit Scoring: What's Inside the Black Box?

By Maria Fernandez Vidal, Jacobo Menajovsky

”

The responsible use of algorithms requires providers to know which variables are being considered in their credit scoring models and how they are affecting people's scores.

Accuracy vs. Interpretability

Random Forest

XGBoost

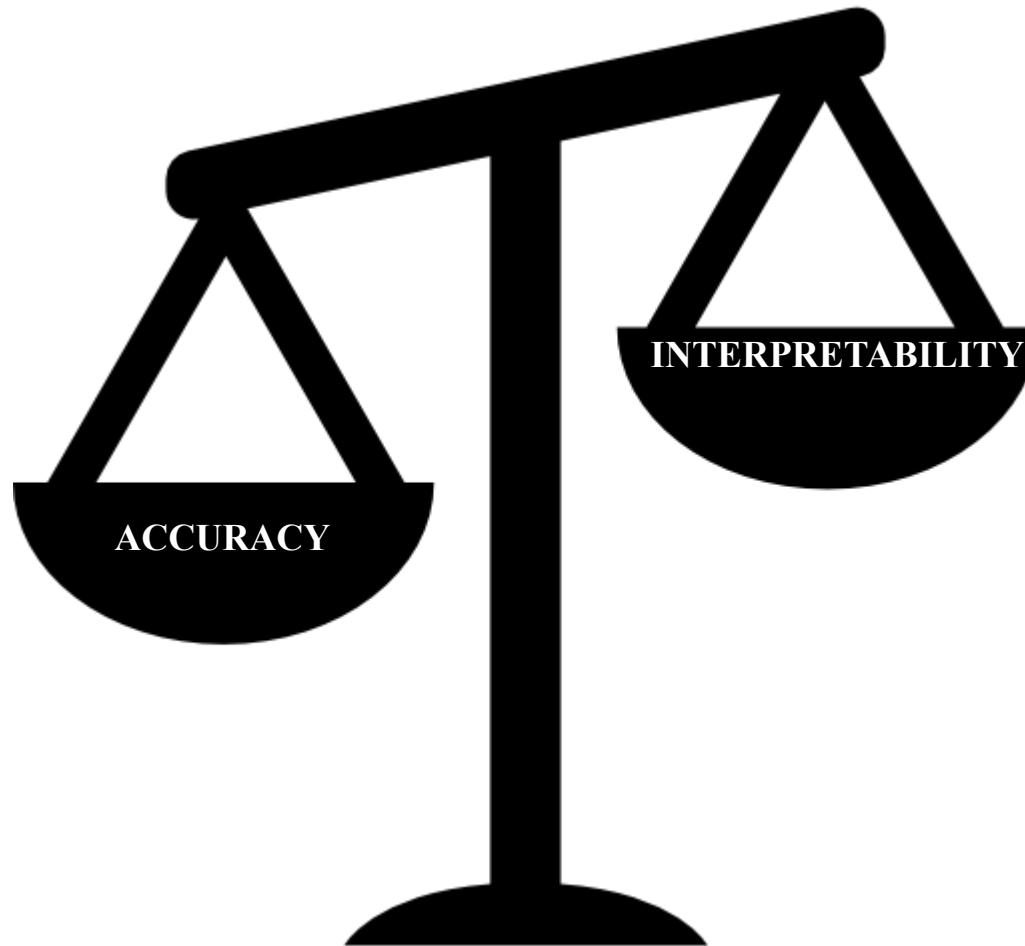
SVM

Deep Learning

...



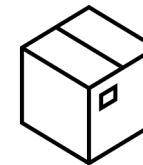
blackbox



Linear Regression

Decision Tree

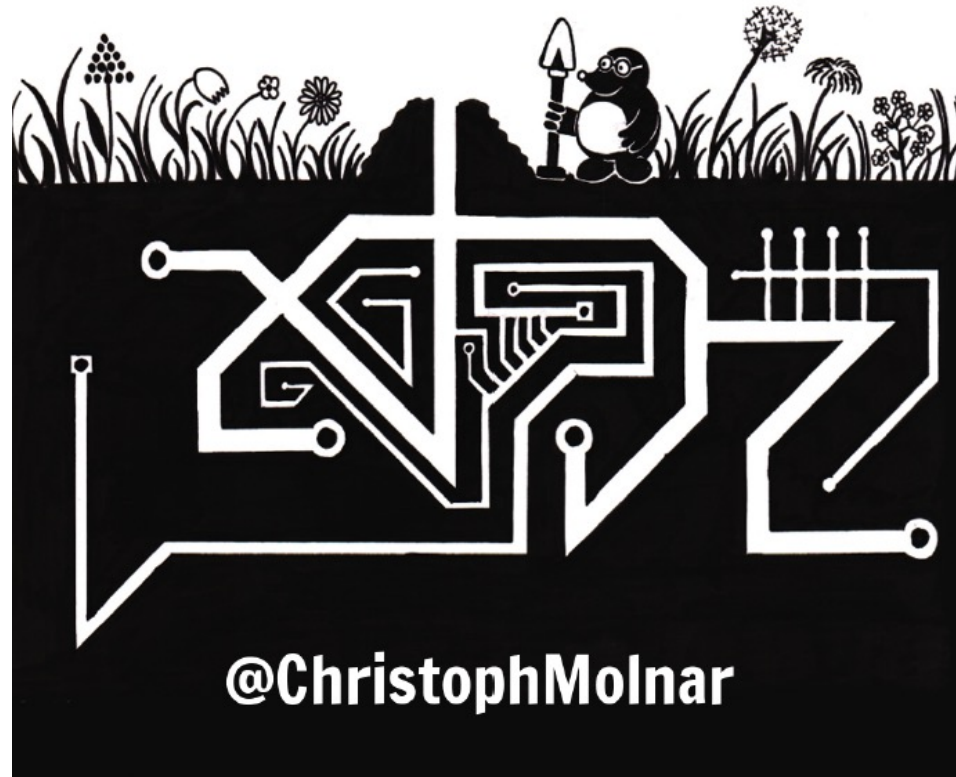
...



**whitebox
(glassbox)**

Interpretable Machine Learning

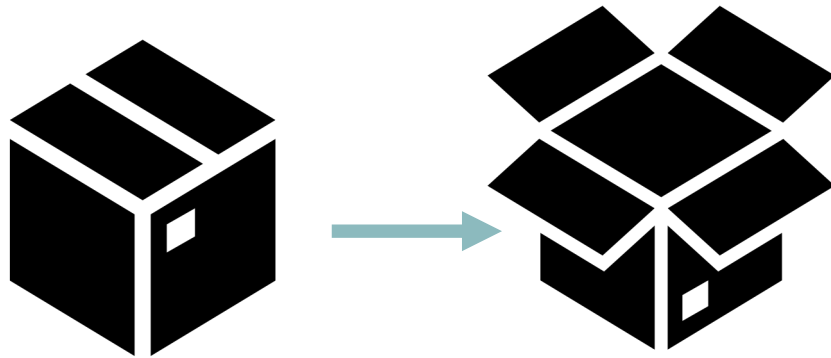
A Guide for Making
Black Box Models Explainable



→ 2019-2021 ...

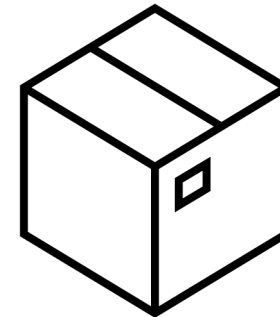
Explainable Artificial Intelligence - XAI

Model Agnostic Explanations



Explaining the blackbox models

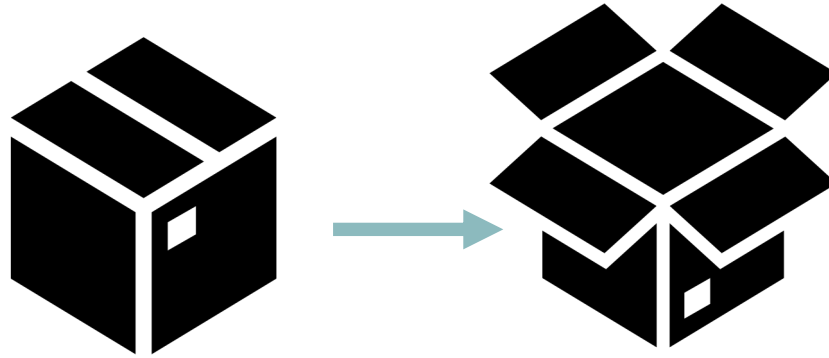
Interpretable Models



High accuracy whitebox models

Explainable Artificial Intelligence - XAI

Model Agnostic Explanations



Explaining the blackbox models

LIME

(RIBERIO VD., 2016)

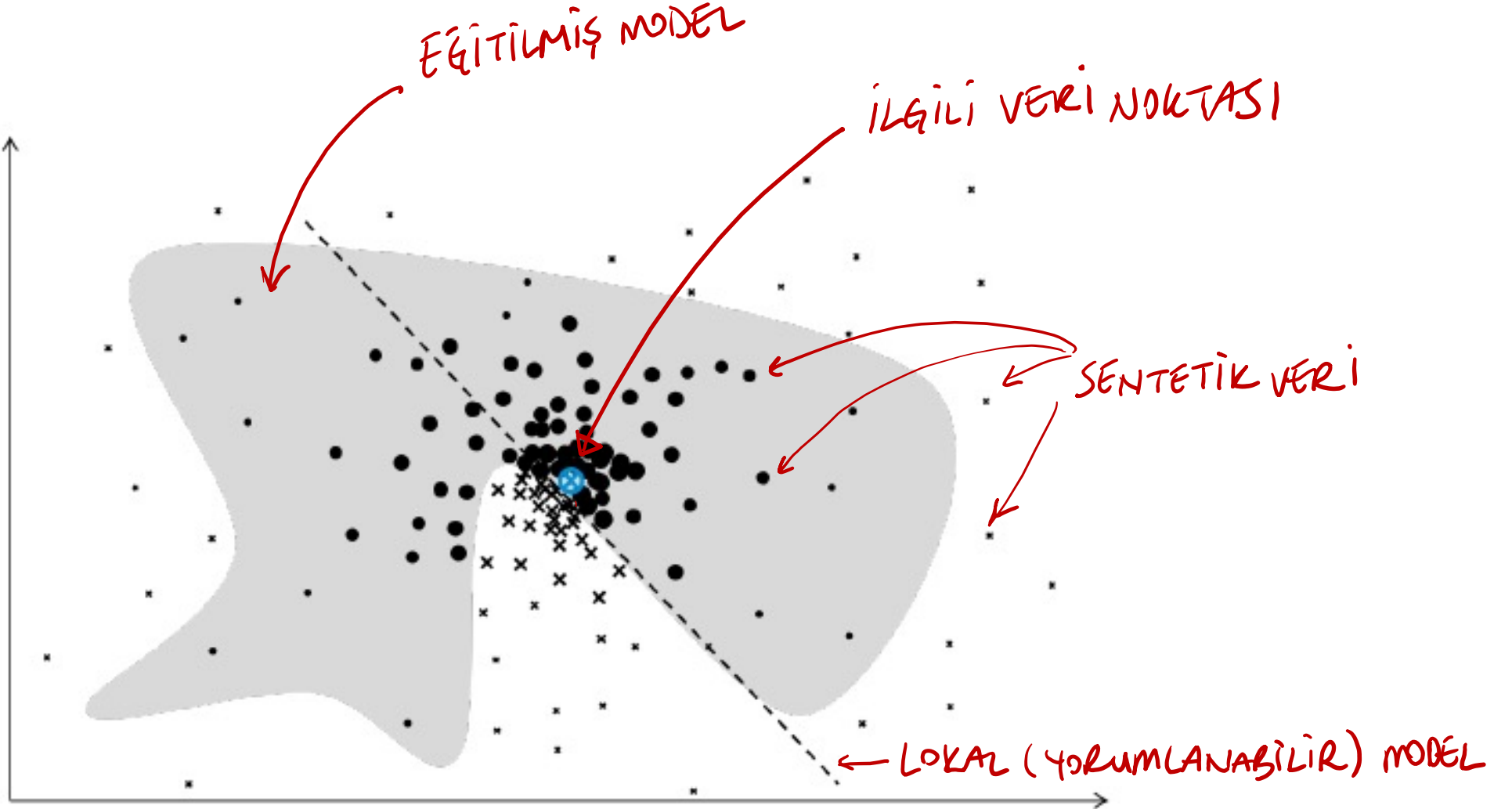
LOCAL INTERPRETABLE MODEL-AGNOSTIC EXPLANATIONS

MODEL
ETİKETLERİ



● × } AĞIRLIKLAR

↑
MESAFE İLE
TERS ORANTILI



LIME

AÇIKLAMA

$$\xi(x) = \arg \min_{g \in G} L(f, g, \pi_x) + \Omega(g)$$

ÖĞRENME HATASI

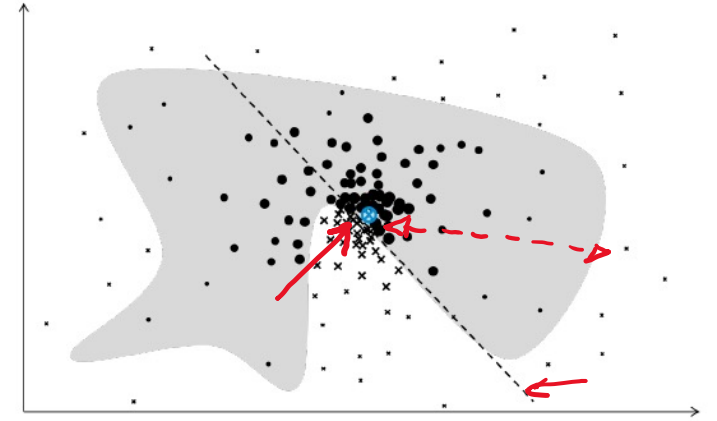
UZAKLIK

İLGİLİ VERİ NOKTASI

LOKAL MODEL

LOKAL MODEL KARMASIKLIĞI

- ÖZGİTİMELİK SAYISI
- AĞAÇ DERİNLİĞİ
- ...



LIME

XGBoost model predicted an acceptance probability of 77.0%

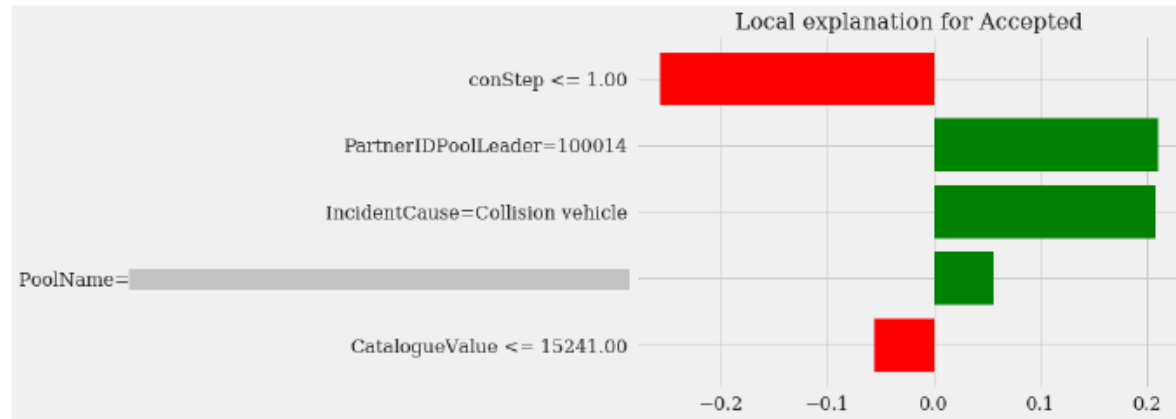


Figure 23: FP - LIME by Default (5 Features)

[Local Pred. = 0.58, Intercept = 0.42, $R^2 = 0.28$, RMSE = 0.26, Time = 4.67 s]

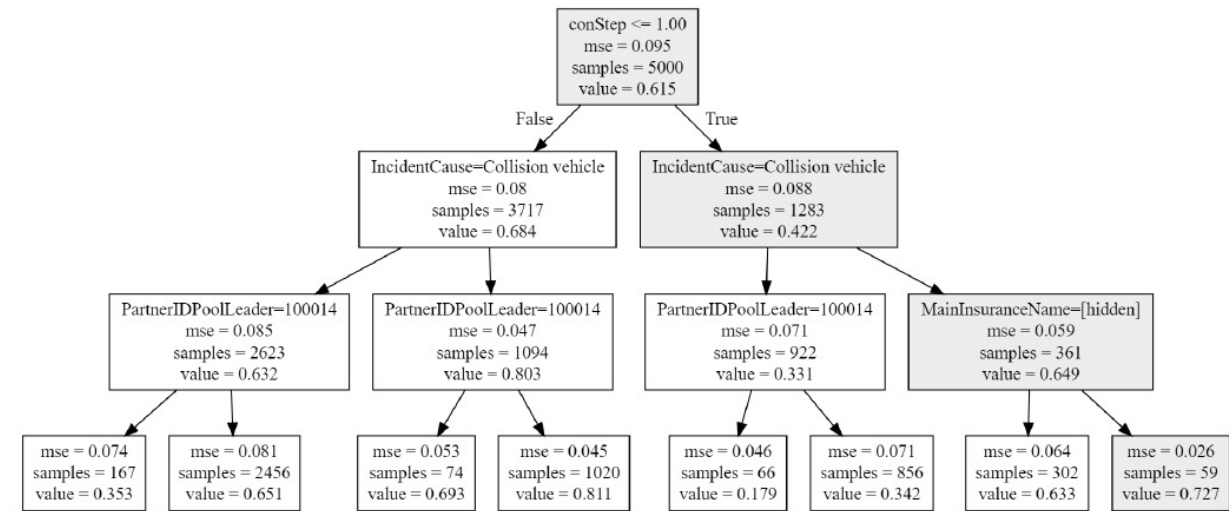


Figure 24: FP - LIME Decision Tree (3 Layers)

[Local Pred. = 0.73, $R^2 = 0.28$, RMSE = 0.26, Time = 2.15 s]

LIME



Even if you **replace the underlying machine learning model**, you can still use the same local, interpretable model for explanation.

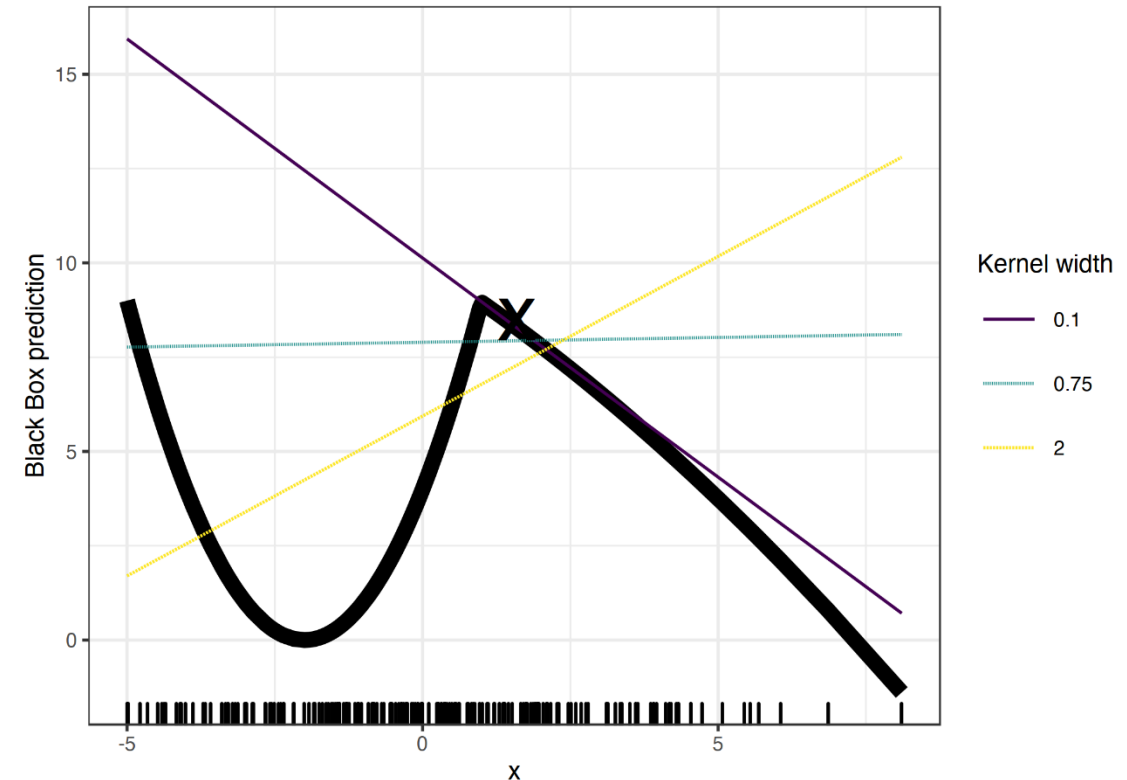
When using Lasso or short trees, the resulting **explanations are short (= selective) and possibly contrastive**.

The default model performs differently when the number of features varies.



LIME explanations, where G is the class of decision trees, yield a different type of explanation.

The correct definition of the neighborhood is a very big, unsolved problem when using LIME with tabular data



SHAP

(LUNDBERG & LEE, 2017)

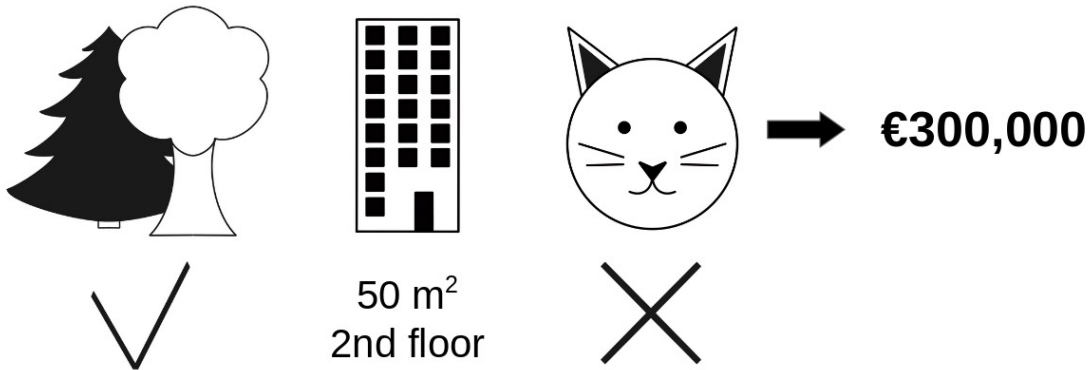
SHAPLEY ADDITIVE EXPLANATIONS

park-nearby + cat-banned + area-50 + floor-2nd

€300,000

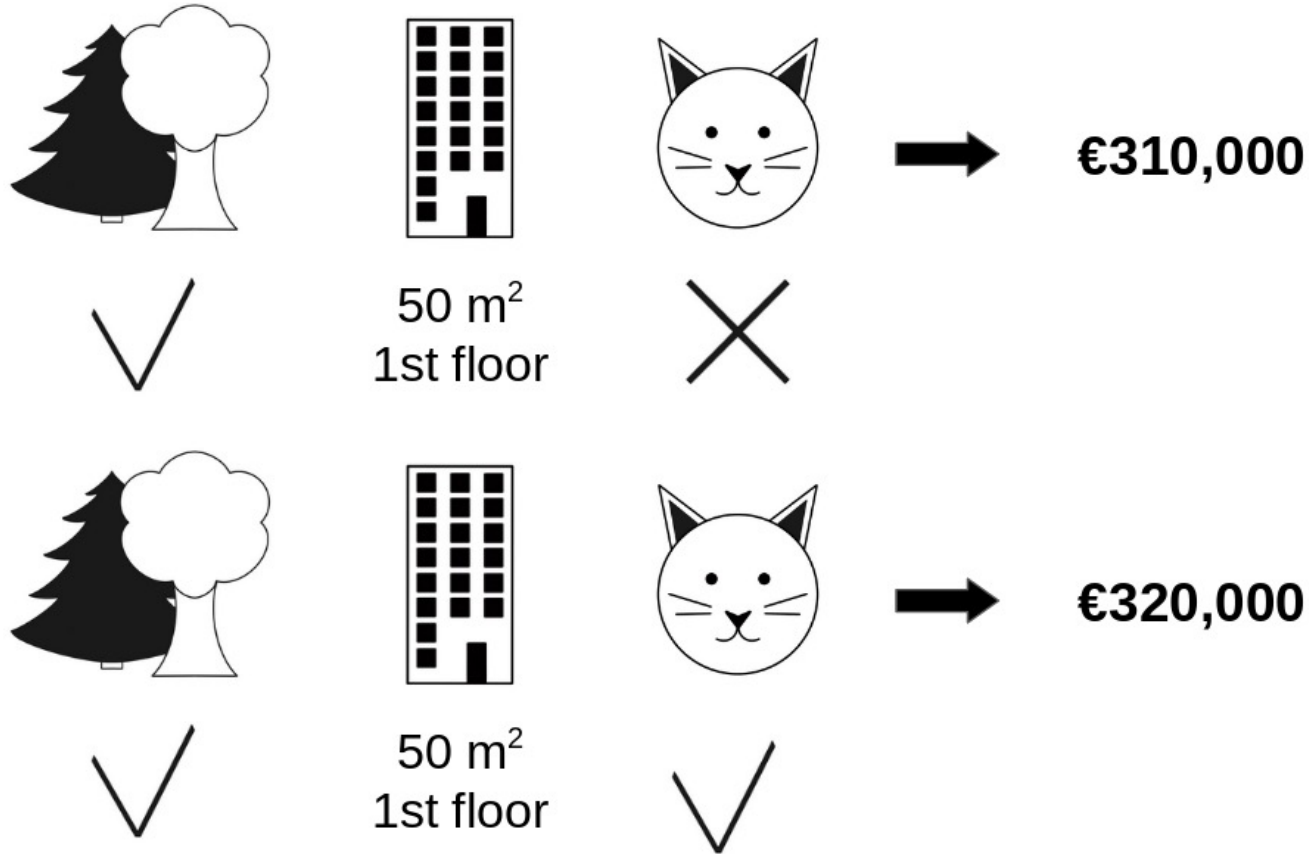
€310,000

€-10,000



SHAP
(LUNDBERG & LEE, 2017)

SHAPLEY ADDITIVE EXPLANATIONS



SHAP

(LUNDBERG & LEE, 2017)

- No feature values
- park-nearby
- area-50
- floor-2nd
- park-nearby + area-50
- park-nearby + floor-2nd
- area-50 + floor-2nd
- park-nearby + area-50 + floor-2nd

SHAPLEY ADDITIVE EXPLANATIONS



✓

50m^2



2nd floor



✓

50m^2



✓



2nd floor

50m^2



2nd floor



✓

50m^2



2nd floor

SHAP

(LUNDBERG VE LEE, 2017)

SHAPLEY ADDITIVE EXPLANATIONS

AÇIKLAMA
MODELİ

$$g_i(\mathbf{z}') = \phi_0 + \sum_{j=1}^M \phi_{ij} z'_{ij}$$

ÖRNEK

ÖZNEİTELİK

ÖZNEİTELİK BU ÖRNEKTE
VAR (1), YOK (0)

SHAPLEY
DEĞERİ

$$\phi_{ij} = \sum_{S \subseteq X_i \setminus \{j\}} \frac{|S|!(M - |S| - 1)!}{M!} [f_i(S \cup \{j\}) - f_i(S)]$$

j HARİÇ
ÖZNEİTELİKLER

MODEL
TAHMİNİ

SHAP



$$\phi_{ij} = \sum_{S \subseteq X_i \setminus \{j\}} \frac{|S|!(M - |S| - 1)!}{M!} [f_i(S \cup \{j\}) - f_i(S)]$$

Tüm
ALT KÜMELER
2^M

YAKLAŞIKLAMA YÖNTEMLERİ

KERNEL SHAP*

LINEAR SHAP

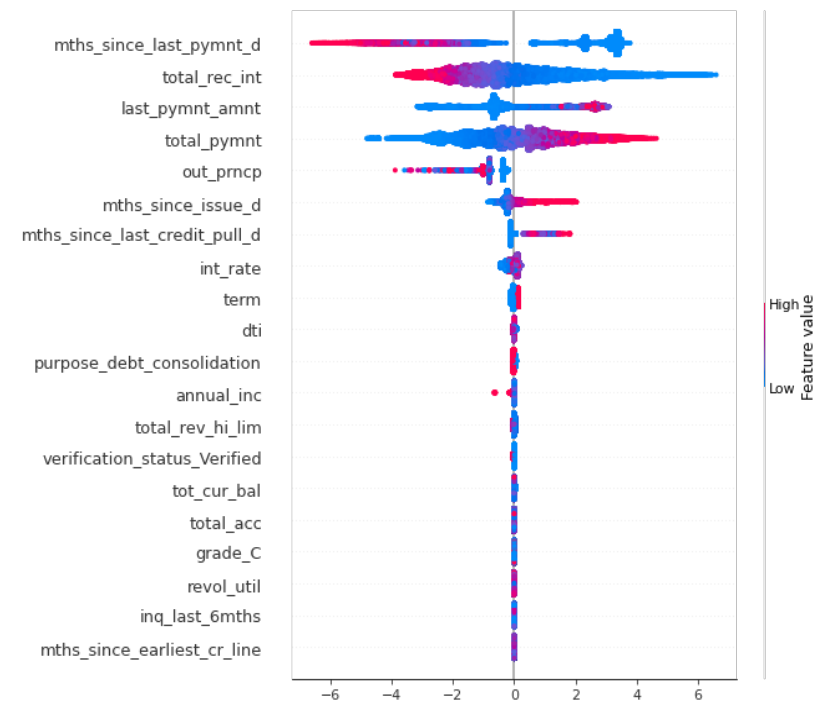
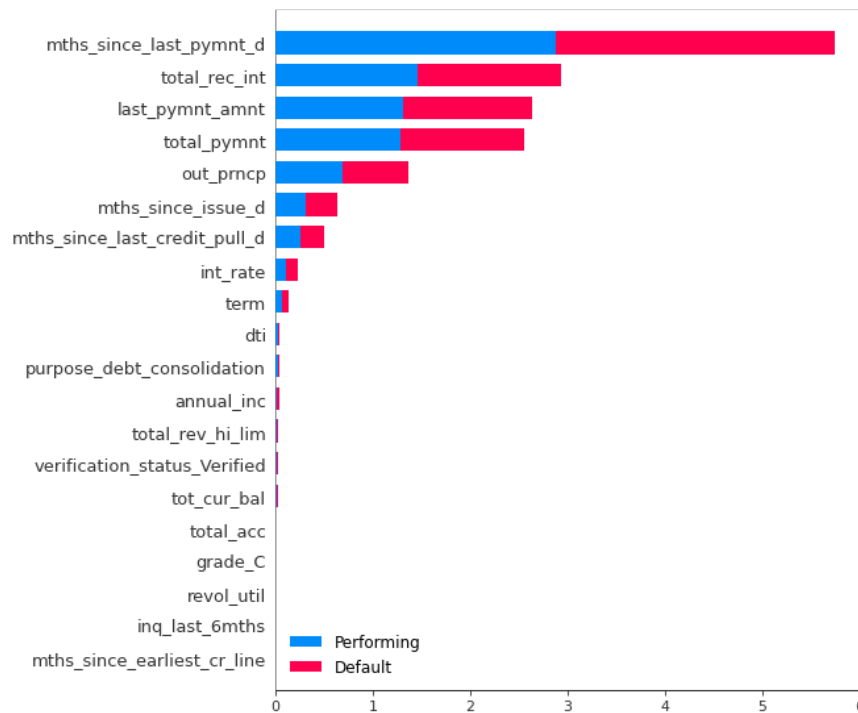
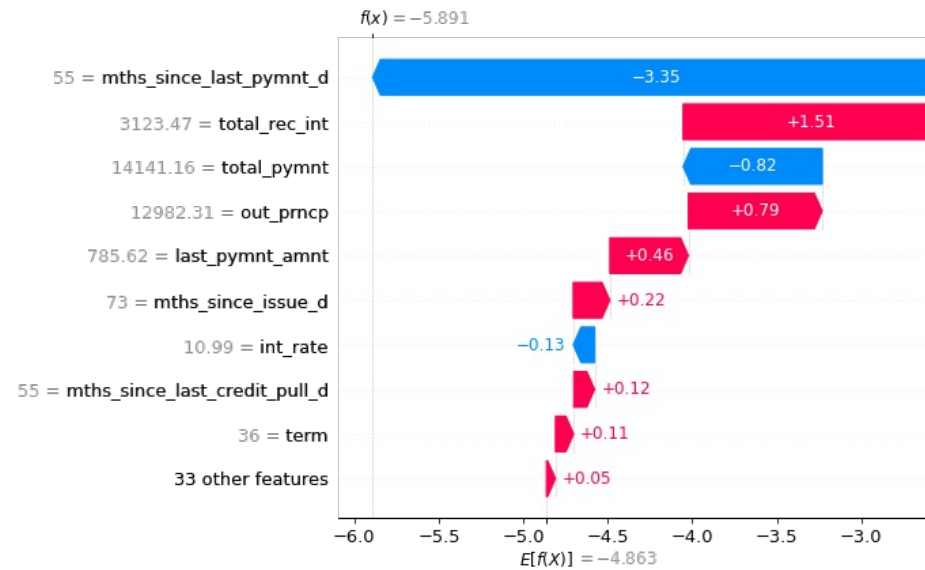
LOW-ORDER SHAP

TREE SHAP

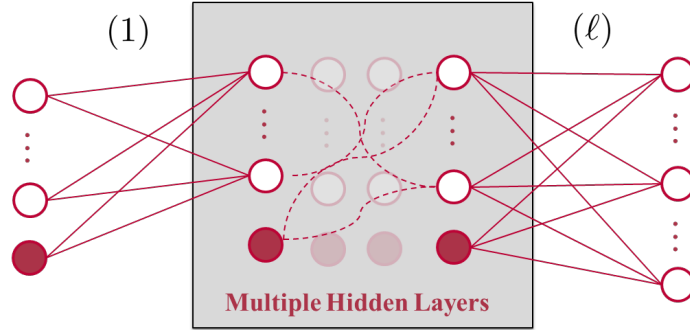
DEEP SHAP

* AGNOSTİK

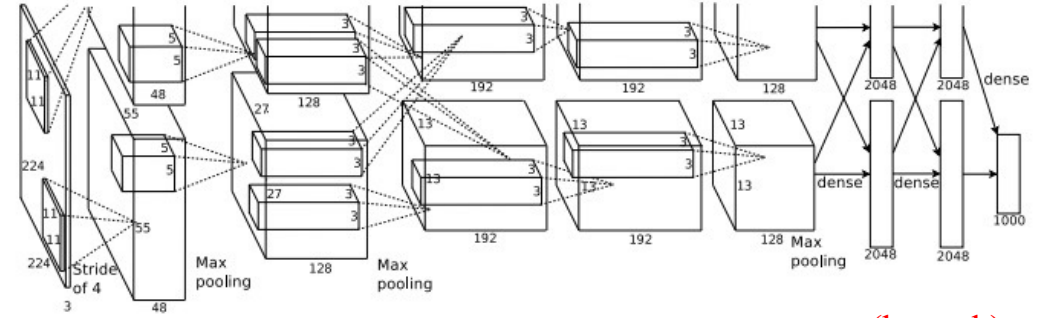
SHAP



DERİN ÖĞRENME



$$\hat{y}_k(X, \beta) = \sigma \left(\sum_j \beta_{kj}^{(\ell)} h \left(\sum_s \beta_{js}^{(\ell-1)} h \left(\dots h \left(\sum_i \beta_{ji}^{(1)} X_i \right) \dots \right) \right) \right)$$



(kaynak)

GÖRÜNTÜ İŞLEME ✓

METİN İŞLEME ~

DİĞER PROBLEMLER ?

LIME

SHAP

InterpretML: A Unified Framework for Machine Learning Interpretability

Harsha Nori
Samuel Jenkins
Paul Koch
Rich Caruana
Microsoft Corporation
1 Microsoft Way
Redmond, WA 98052, USA

HANORI@MICROSOFT.COM
SAJENKIN@MICROSOFT.COM
PAULKOCH@MICROSOFT.COM
RCARUANA@MICROSOFT.COM



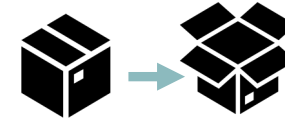
**Understand Models.
Build Responsibly.**

A toolkit to help understand models and enable responsible machine learning

Supported Techniques

Interpretability Technique	Type	Examples
Explainable Boosting	glassbox model	Notebooks
Decision Tree	glassbox model	Notebooks
Decision Rule List	glassbox model	Coming Soon
Linear/Logistic Regression	glassbox model	Notebooks
SHAP Kernel Explainer	blackbox explainer	Notebooks
SHAP Tree Explainer	blackbox explainer	Coming Soon
LIME	blackbox explainer	Notebooks
Morris Sensitivity Analysis	blackbox explainer	Notebooks
Partial Dependence	blackbox explainer	Notebooks

Counterfactual Explanations



Explanation in artificial intelligence: Insights from the social sciences

Tim Miller

School of Computing and Information Systems, University of Melbourne, Melbourne, Australia

Artificial Intelligence, 2019

COUNTERFACTUAL EXPLANATIONS WITHOUT OPENING THE BLACK BOX: AUTOMATED DECISIONS AND THE GDPR

Sandra Wachter,^{*} Brent Mittelstadt,^{**} & Chris Russell^{***}

Harvard Journal of Law & Technology, 2018

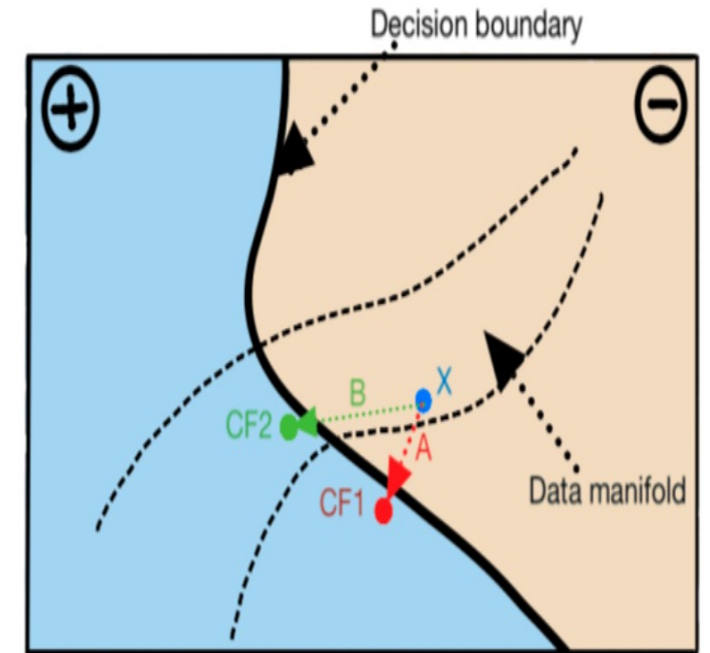
Counterfactual Explanations for Machine Learning: A Review

Sahil Verma
University of Washington
Arthur AI
vsahil@cs.washington.edu

John Dickerson
Arthur AI
University of Maryland
john@arthur.ai

Keegan Hines
Arthur AI
keegan@arthur.ai

arXiv, 2020



(Verma et al., 2020)

Factual Instance

Beatrice is 27yo
Full-time job: 45000 \$/y
Account balance: 50000\$



Counterfactual

Beatrice is 27yo
Full-time job: **50000** \$/y
Account balance: **60000**\$

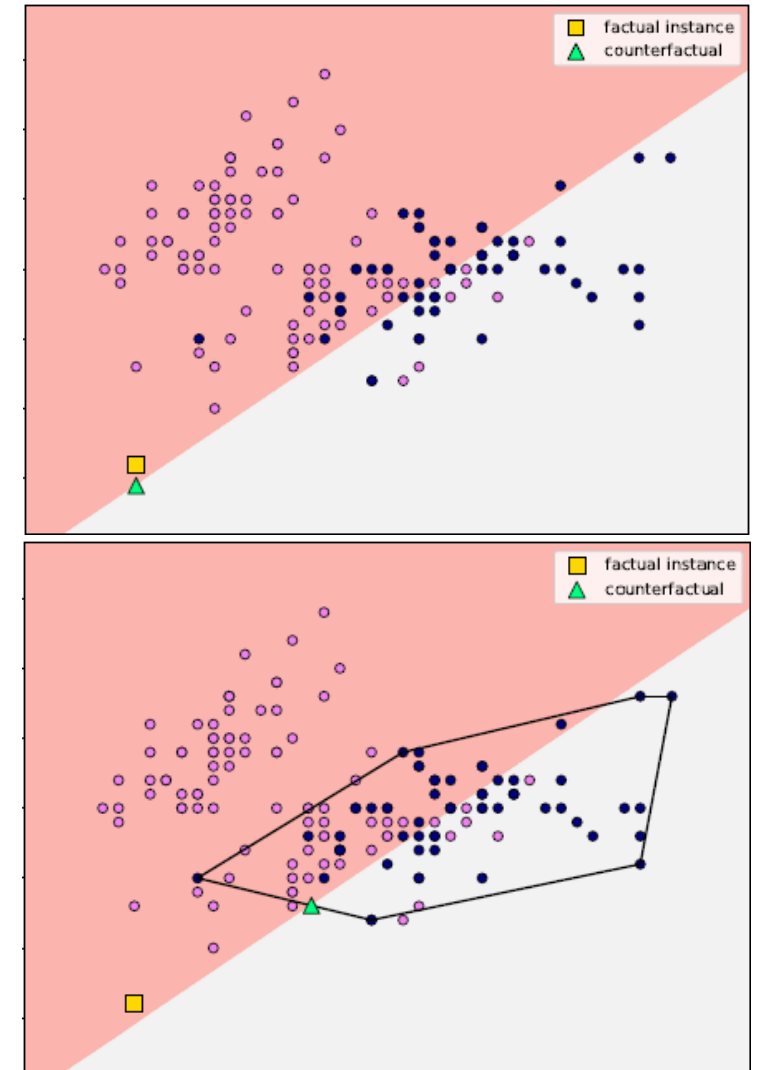
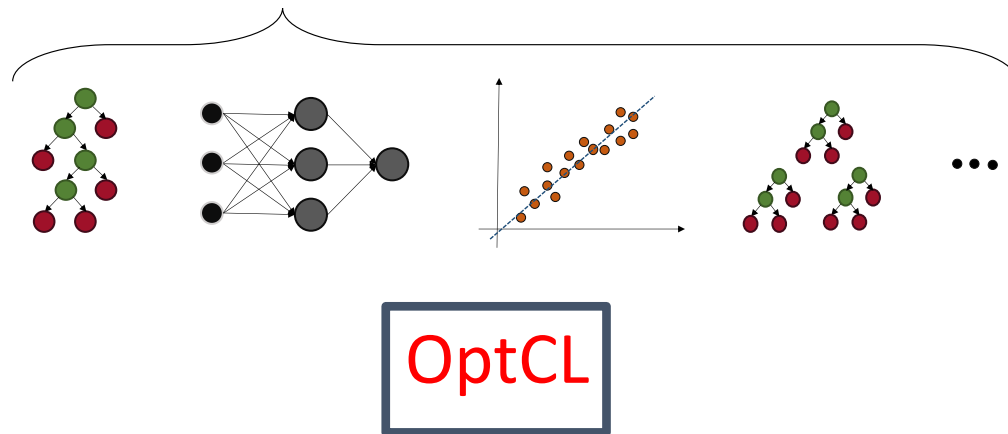


Counterfactual: set of features that should be changed in order to flip a model's prediction

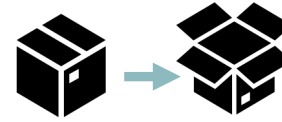
Optimization with Constraint Learning (Birbil et al. 2021)

$$h(\bar{x}) = \bar{y}$$

$$\begin{array}{ll} \underset{\mathbf{x} \in \mathbb{R}^d}{\text{minimize}} & d(\bar{x}, \mathbf{x}) \\ \text{subject to} & h(\mathbf{x}) = y \end{array}$$



Counterfactual Explanations



DHH ✓
@dhh

...

The @AppleCard is such a **** sexist program. My wife and I filed joint tax returns, live in a community-property state, and have been married for a long time. Yet Apple's black box algorithm thinks I deserve 20x the credit limit she does. No appeals work.

[Traduci il Tweet](#)

9:34 PM · 7 nov 2019 · Twitter for iPhone



Apple Card ✓
@AppleCard

Well, if your wife would have had a 20+ years relationship with our bank, and would have been regarded as Premium customer at some point in time, she would also receive a 20x credit limit.

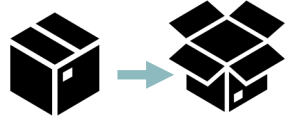


Apple Card ✓
@AppleCard

Well, if your wife's relationship status would have been "husband" instead of "wife", she would also receive a 20x credit limit.

We clearly messed up, we are updating our models now.

CE Case Study

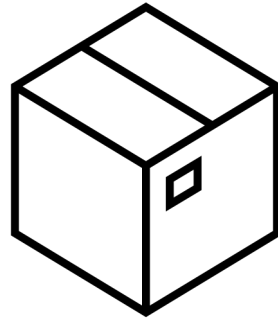


y : good or bad credit risk?

Label	Variable name	Description
F1	duration	Duration in months
F2	credit_amount	Credit amount
F3	instalment_commitment	Installment rate in percentage of disposable income
F4	age	Age in years
F5	residence_since	Present residence since X years
F6	existing_credits	Number of existing credits at this bank
F7	num_dependents	Number of people being liable to provide maintenance for
F8	checking_status	Status of existing checking account, in Deutsche Mark
F9	credit_history	Credit history (credits taken, paid back duly, delays, critical accounts)
F10	employment	Present employment, in number of years.
F11	foreign_worker	Foreign worker (yes,no)
F12	housing	Housing (rent, own,...)
F13	job	Job
F14	other_parties	Other debtors / guarantors
F15	other_payment_plans	Other installment plans (banks, stores)
F16	own_telephone	Telephone (yes,no)
F17	personal_status	Personal status (married, single,...) and sex
F18	property_magnitude	Property (e.g. real estate)
F19	purpose	Purpose of the credit (car, television,...)
F20	saving_status	Status of savings account/bonds, in Deutsche Mark.

	F1	F2	F3	F4	F8*	F10	F12	F14	F16	F18*	F20*
$\hat{x}$	24.0	1371.26	4.0	25.0	A	$1 \leq X < 4$	rent	none	none	A	A
Part A: validity, proximity, coherence											
(a)	15.02	-333.52	3.86	27.04	-	-	-	-	-	-	-
Part B: validity, proximity, coherence, sparsity											
(a)	7.12	-	-	-	-	-	-	-	-	-	-
Part C: validity, proximity, coherence, sparsity, diversity											
(a)	7.12	-	-	-	-	-	-	-	-	-	-
(b)	-	-2873.47	-	30.06	-	-	-	-	-	-	-
(c)	-	-	1.96	26.63	-	-	-	-	-	-	-
Part D: validity, proximity, coherence, sparsity, diversity, actionability											
(a)	7.12	-	-	-	-	-	-	-	-	-	-
(b)	-	-	1.96	26.63	-	-	-	-	-	-	-
(c)	-	-	-	75.52	-	-	-	-	-	-	-
Part E: validity, proximity, coherence, sparsity, diversity, actionability, data manifold closeness											
(a)	22.0	1283.52	-	-	B	$4 \leq X < 7$	-	-	-	B	-
(b)	10	1363.43	2.0	64.0	B	-	own	-	yes	C	B
(c)	12.0	1893.04	-	29.0	-	-	own	guarantor	yes	B	B
Part F: validity, proximity, coherence, sparsity, diversity, actionability, data manifold closeness, causality											
(a)	-	-	-	-	B	$4 \leq X < 7$	-	-	-	B	-
(b)	22.0	990.51	-	-	B	$4 \leq X < 7$	-	-	-	B	-
(c)	26.83	1910.28	-	-	B	$4 \leq X < 7$	-	-	-	B	-

Interpretable Models

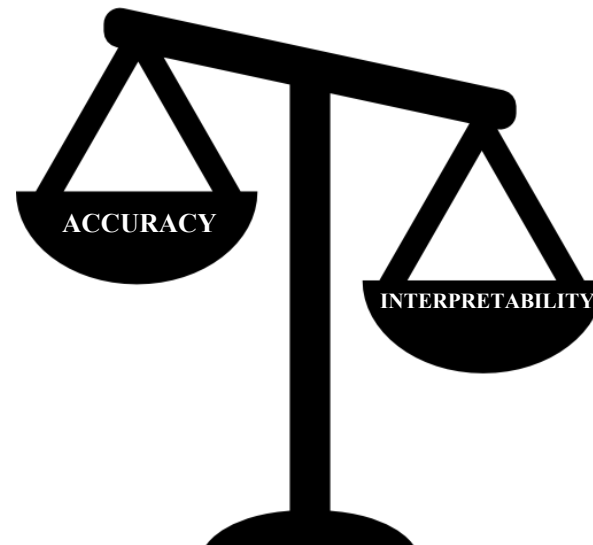
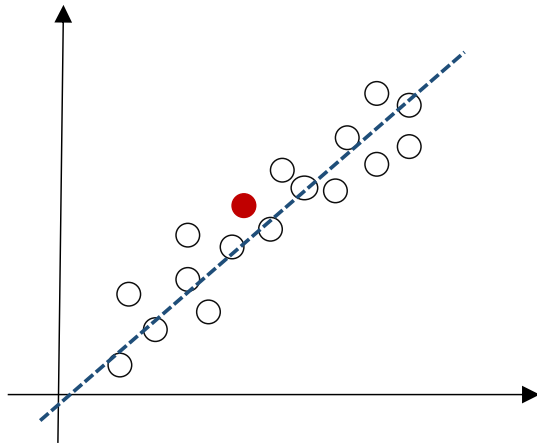


High accuracy whitebox models

Linear Regression

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p$$

$$\hat{y} = 1.0 + 2.0x_1 - 3.0x_2$$



EBM

(NORI vd., 2019)

EXPLAINABLE BOOSTING MACHINE

“ÖRNEK i İÇİN TAHMİN

$$g(E[y_i]) = \beta_0 + \sum_{j=1}^M f_j(x_{ij}) + \sum_{j=1}^M \sum_{k \neq j} f_{jk}(x_{ij}, x_{ik})$$

ÖZNİTELİKLER ARASI
İLİŞKİLER

GENELEŞTİRİLMİŞ
DOĞRUSAL
MODEL

LINK
FONKSİYONU
(ÖRN. LOGIT)

ÖZNİTELİK
FONKSİYONLARI

GRADIENT
BOOSTING
MODELLERİ

EBM

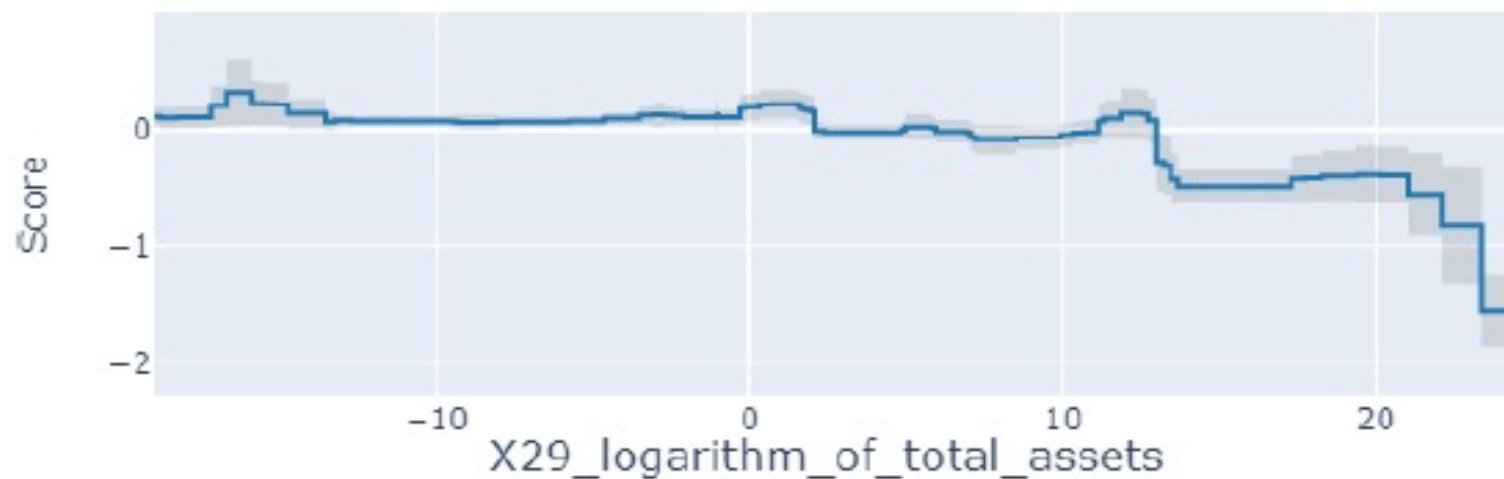


Figure 1: GAM plot of the effect of a feature $\log(\text{total assets})$ on the prediction of bankruptcy for a data set of Polish firms. There is quite some variance in the relation, but overall higher assets clearly decrease the bankruptcy risk as would be expected.

SLIM

(üstün vd., 2013)

SUPERSPARSE LINEAR INTEGER MODEL

HATA

$$\min_{\lambda} \frac{1}{N} \sum_{i=1}^N \mathbb{1}[y_i x_i^T \lambda \leq 0] + C_0 \|\lambda\|_0 + C_1 \|\lambda\|_1,$$

s.t. $\lambda \in \mathcal{L}$

$\{-1, 1\}$

YORUMLANABİLİRLİK
(AÇI SAHIDA ÖZGÜTEK)

SLIM

PREDICT MUSHROOM IS POISONOUS IF SCORE > 3				
1.	<i>spore_print_color = green</i>	4 points		
2.	<i>stalk_surface_above_ring = grooves</i>	2 points	+	
3.	<i>population = clustered</i>	2 points	+	
4.	<i>gill_size = broad</i>	-2 points	+	
5.	<i>odor</i> $\in \{none, almond, anise\}$	-4 points	+	
ADD POINTS FROM ROWS 1-5		SCORE	=	

Figure 2: The scoring system for mushroom edibility produced by SLIM as displayed in Ustun and Rudin (2015)

ANKET

14 VERİ ANALİZİ UZMANI
(11 KİŞİ YAPAY ÖĞRENME ALANINDA)

Table 9: Average scores of survey respondents on model specific questions as well as corresponding standard deviations (SD). Highest scores in bold.

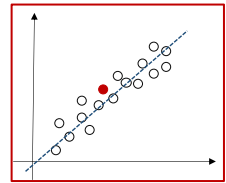
	Familiarity		Test	Understanding		Stakeholders		Practical Value	
	(1-3)	SD	% Correct	(1-7)	SD	(1-7)	SD	(1-7)	SD
Logit	3.00	0.00	78	5.21	1.25	4.57	1.50	5.36	0.93
SLIM	1.36	0.63	64	5.79	1.53	5.71	1.49	4.86	1.75
EBM	1.79	0.70	86	4.64	1.55	4.43	1.79	5.07	1.21
SHAP	2.36	0.63	71	5.43	1.09	5.00	1.24	5.86	0.77

→
XGBOOST +

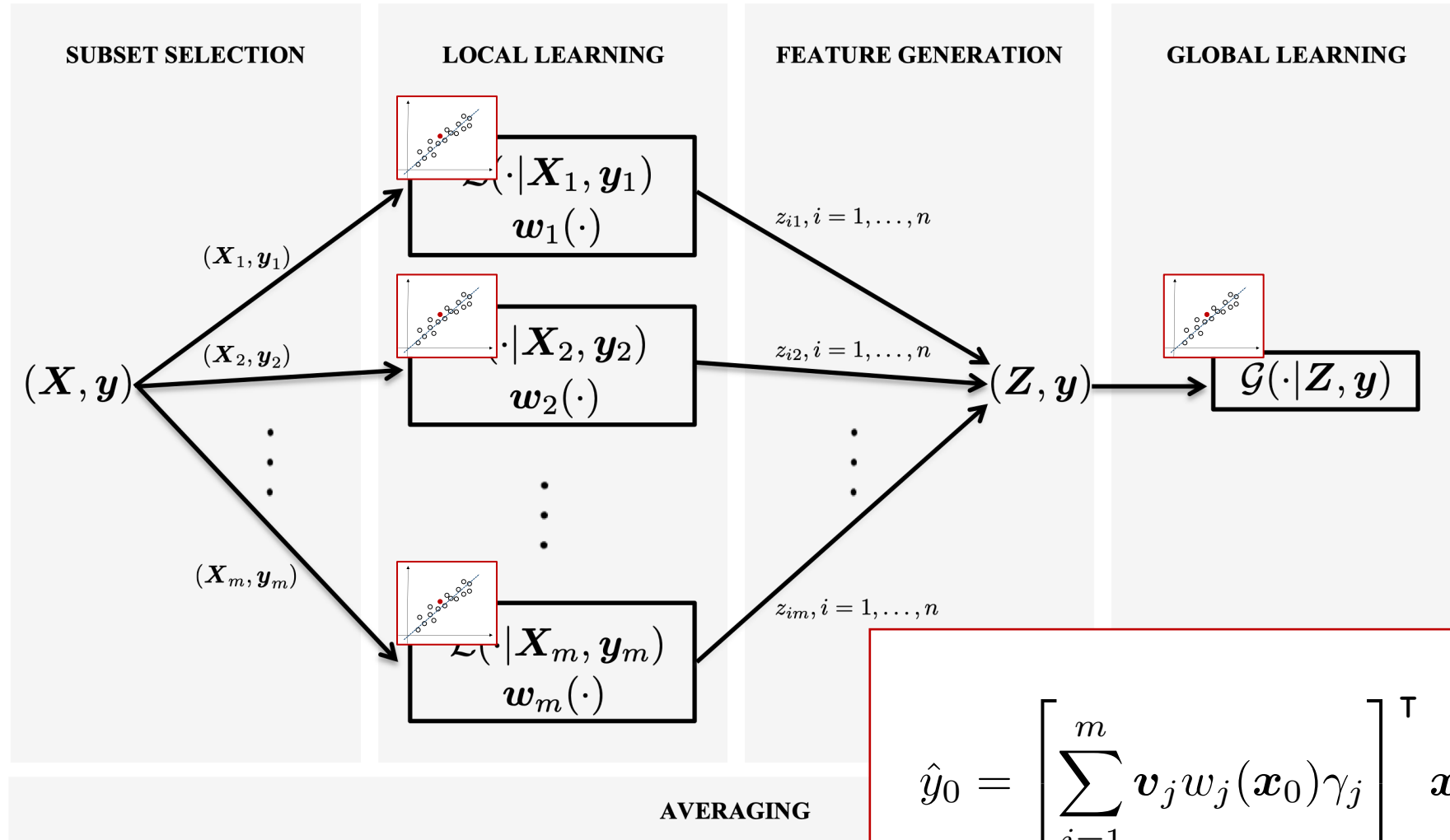
↑
YÖNTEMLE İLGİLİ
DOĞRU-YANLIŞ SORULARI

■ YÖNTEMİ ANLAMA (KİŞİSEL)
■ YÖNTEMİ ANLATMA
■ PRATİKTE KULLANIM ŞANSI

Interpretable LESS (Birbil et al. 2021)



Linear Regression
Lasso
Elastic Net
...

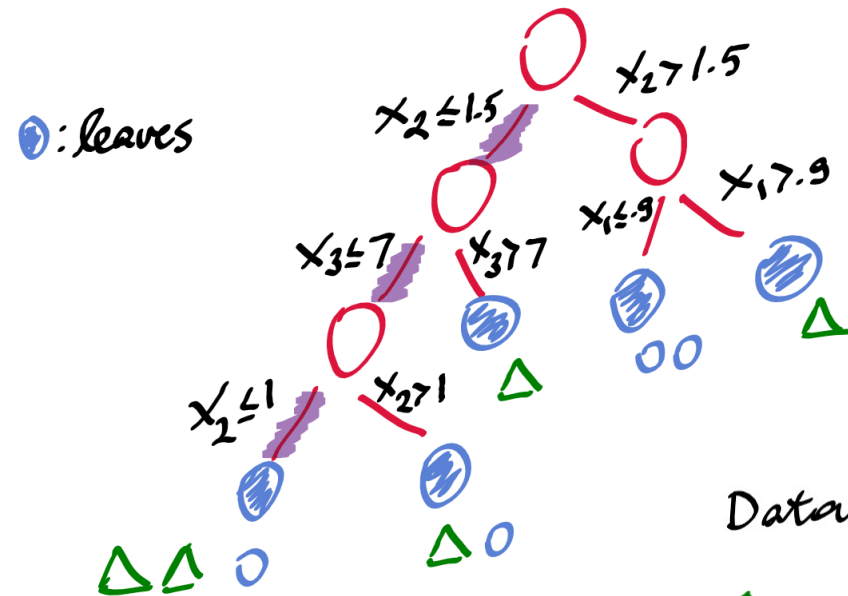


$$\hat{y}_0 = \left[\sum_{j=1}^m v_j w_j(x_0) \gamma_j \right]^\top \quad x_0 = \beta(x_0)^\top x_0$$

RULE-BASED METHODS USING OPTIMIZATION

IF AGE > 25 AND PASTCREDIT=TRUE THEN ACCEPT

DECISION TREES



Rule j : $x_2 \leq 1 \& x_3 \leq 7$
(leaf)

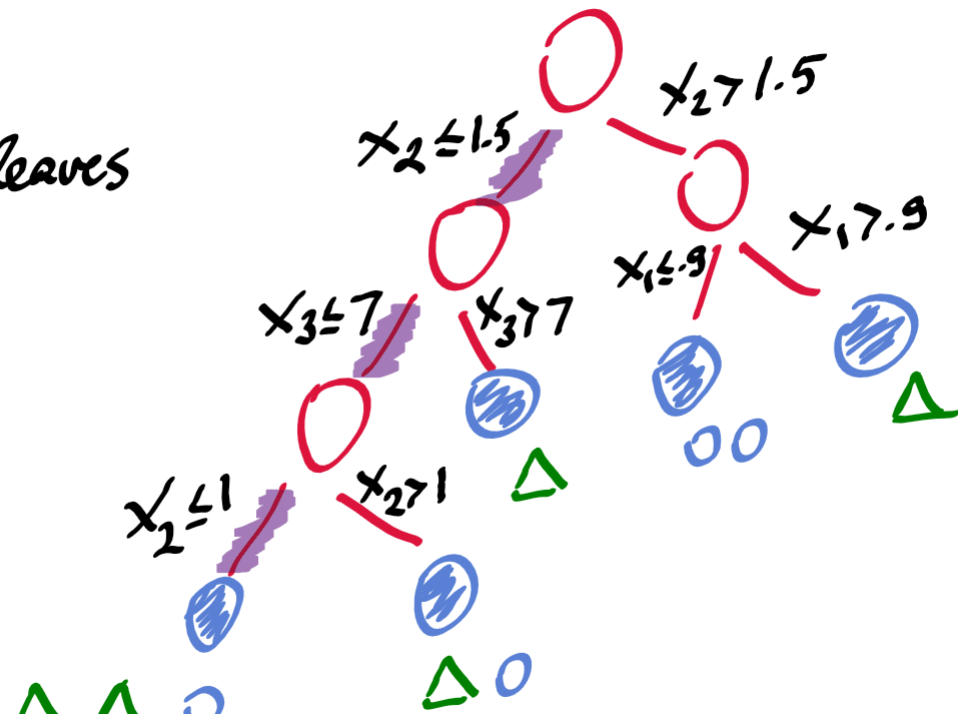
Data: $\{\triangle, \triangle, \circ, \triangle, \circ, \circ\}$

$\triangle \leftarrow$ labels
 $\circ \leftarrow$

Data: $\{\Delta, \Delta, 0, \Delta, 0, 0\}$

 ← Labels
 ←

: leaves



$\Delta \Delta \circ$

$$R_j(x_i) = \Delta$$

Out of sample $< \text{ }_0^{\Delta}$?

- ① Check the rules (leaves)
- ② Classify with majority voting

DATA
 $(x_i, y_i), i \in \underline{I}$

$$x_i = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix}$$

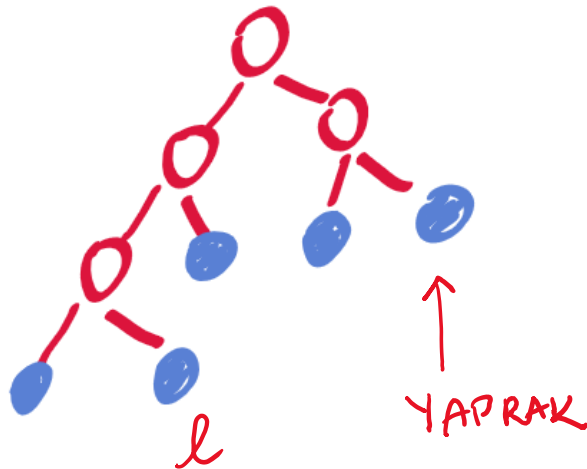
$$x_i \in \mathbb{R}^p, y_i \in \{-1, 1\}$$

OCT

(BERTSIMAS & DUNN, 2017)

(BERTSIMAS & DUNN, 2017)

(x_i, y_i)
 $\uparrow \quad \uparrow$
 GİRİŞ ETİKET



OPTIMAL CLASSIFICATION TREES

SINIFLANDIRMA | HATASI

- AĞAC BÜYÜKLÜĞÜ
(DÜĞÜM SATISI)

$$\min R_{xy}(T) + \alpha|T|$$

$$\text{s.t. } N_x(l) \geq N_{\min}$$

2. YAPRAĞINDAKİ ÖRNEK SAYISI

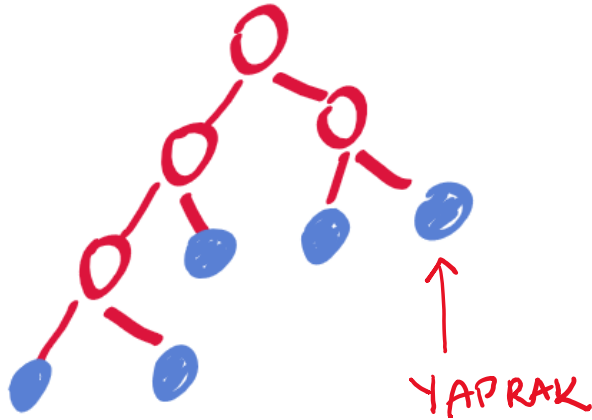
OSDT

(HU VD., 2020)

OPTIMAL SPARSE DECISION TREES

(x_i, y_i)

$\uparrow$
 $\{0,1\}$



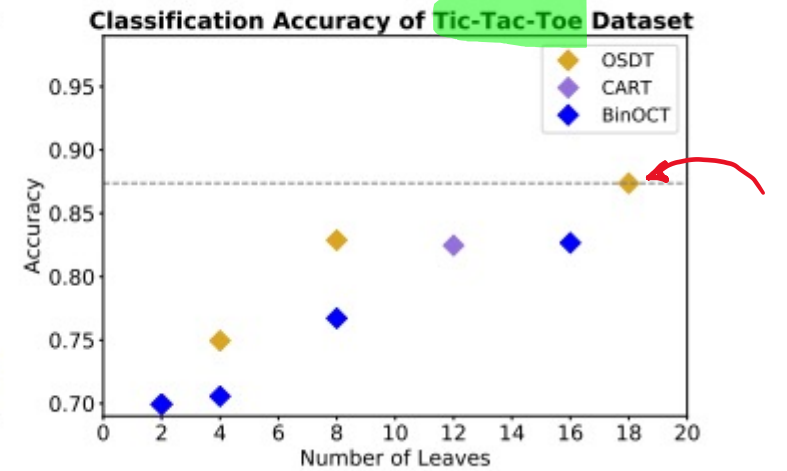
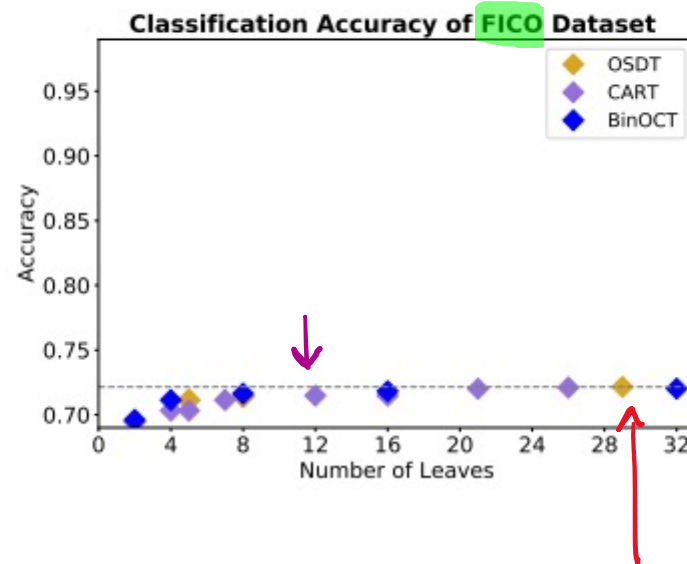
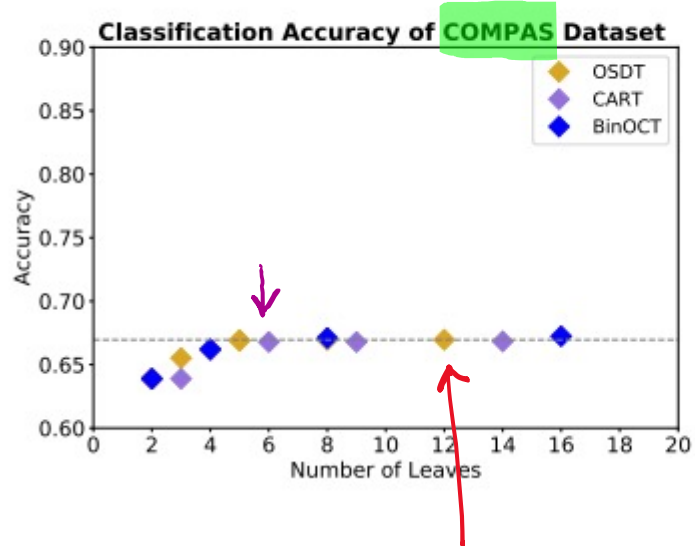
$$R(d, \mathbf{x}, \mathbf{y}) = \ell(d, \mathbf{x}, \mathbf{y}) + \lambda H_d$$

$\uparrow$
AĞAÇ

$\underbrace{\ell(d, \mathbf{x}, \mathbf{y})}_{\text{SINIFLANDIRMA HATASI}}$

$\underbrace{\lambda H_d}_{\text{AĞAÇTAKİ YAPRAK SAYISI}}$

OSDT



SÜRE LİMİTİ 30 DK.

OSDT

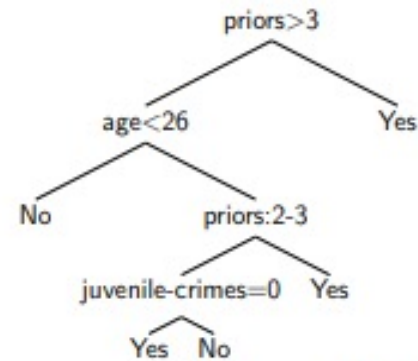


Figure 4: An optimal decision tree generated by OSDT on the COMPAS dataset. ($\lambda = 0.005$, accuracy: 66.90%)

Two Petty Theft Arrests

VERNON PRATER	BRISHA BORDEN
LOW RISK	HIGH RISK
3	8

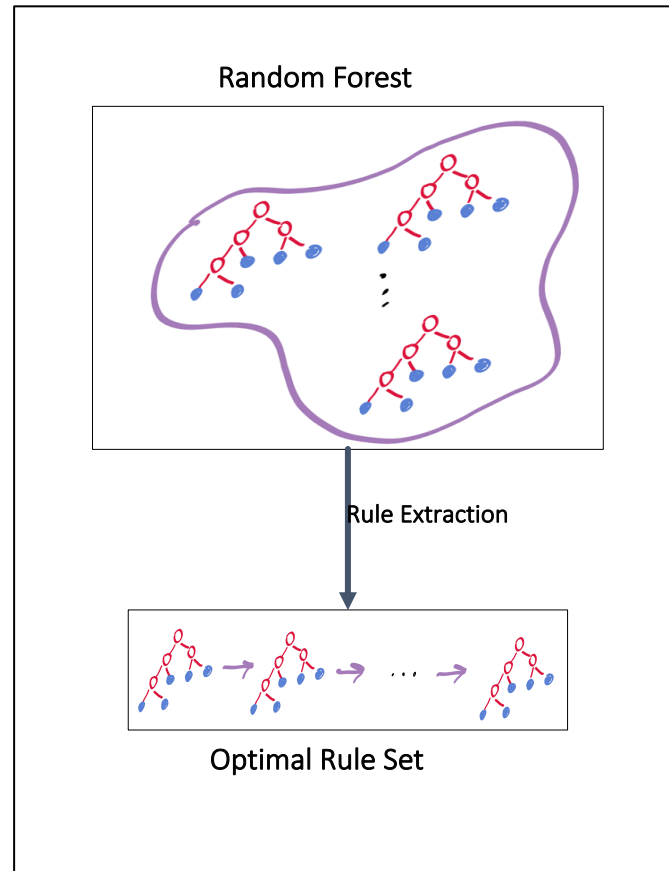
Borden was rated high risk for future crime after she and a friend took a kid's bike and scooter that were sitting outside. She did not reoffend.

VERNON PRATER	BRISHA BORDEN
Prior Offenses 2 armed robberies, 1 attempted armed robbery	Prior Offenses 4 juvenile misdemeanors
Subsequent Offenses 1 grand theft	Subsequent Offenses None

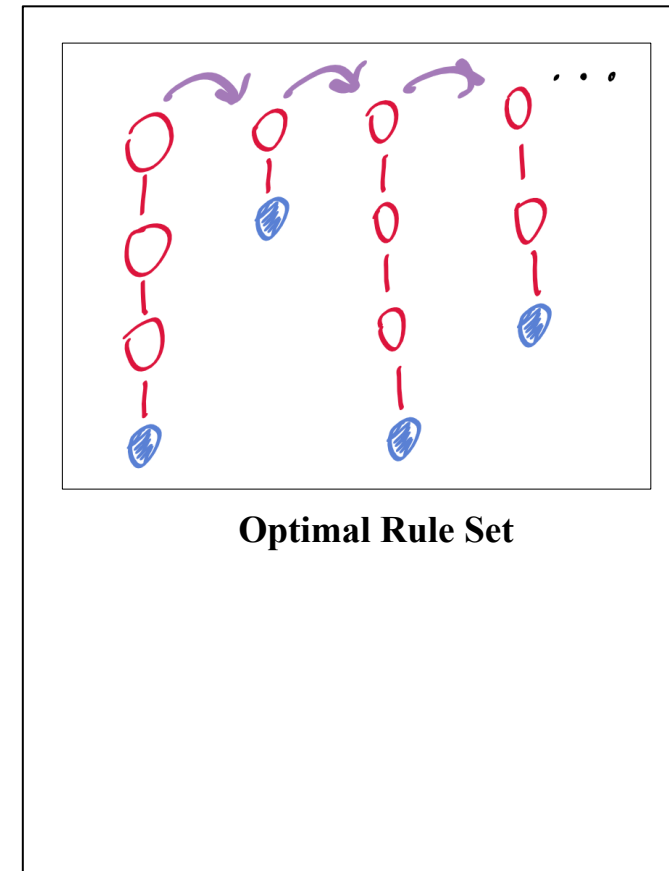
Optimal Rule Discovery

(Akyüz, Birbil 2021)

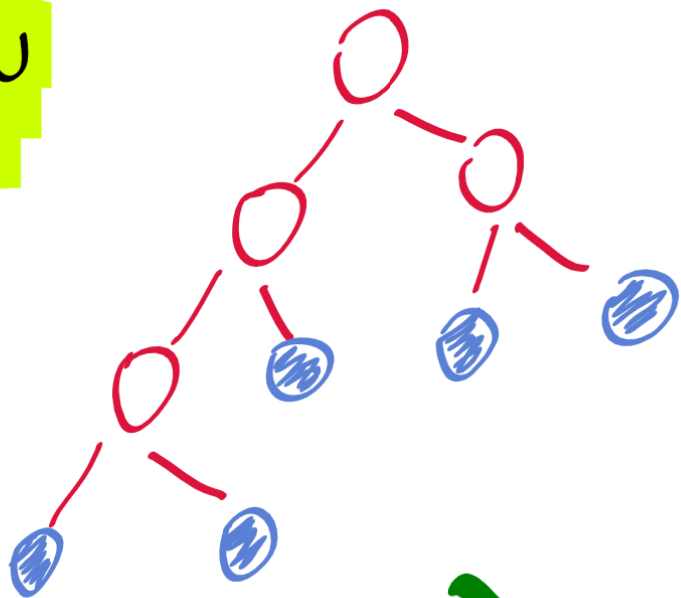
Rule Extraction



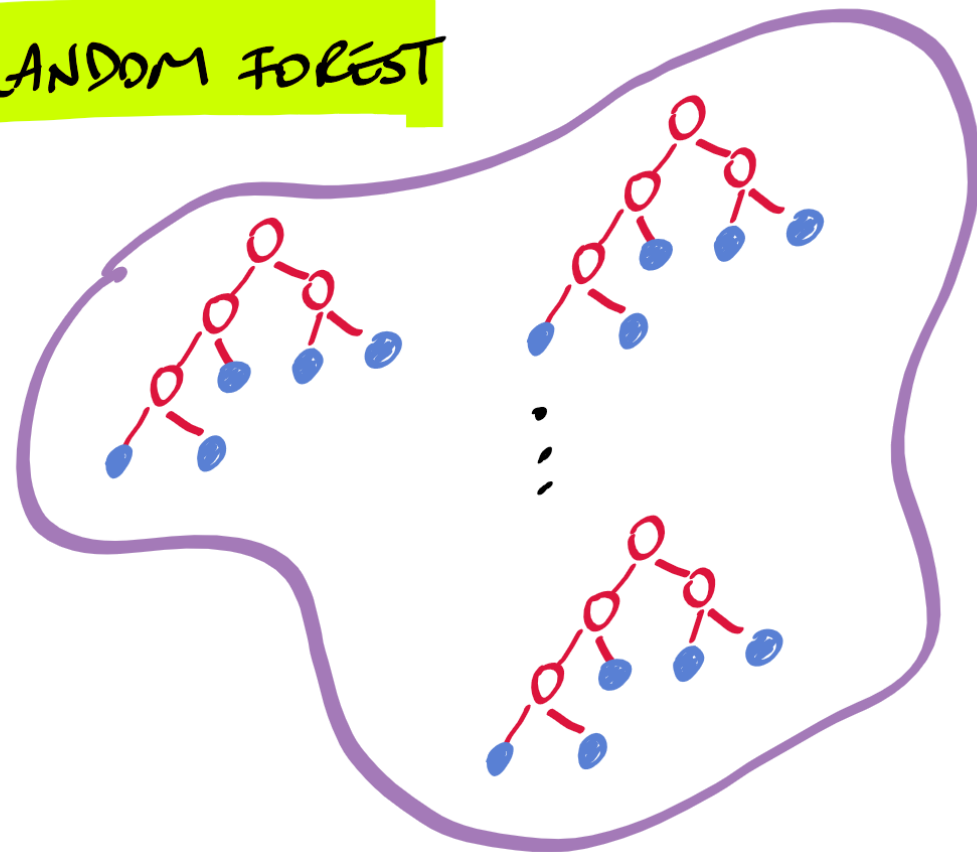
Rule Generation



DECISION TREES



RANDOM FOREST



VARIANCE ↓ ACCURACY ↑

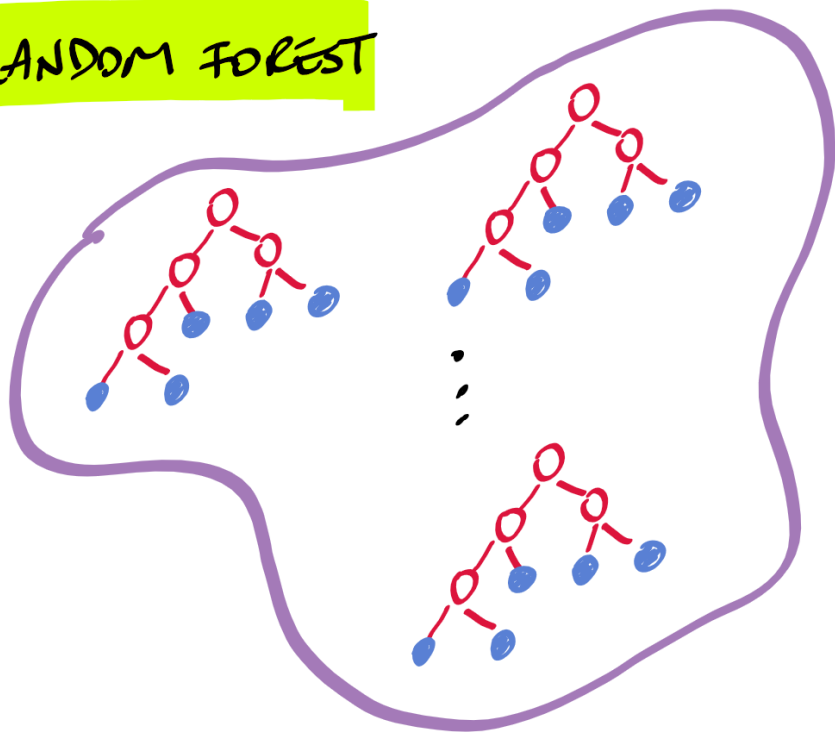


INTERPRETABILITY ↓



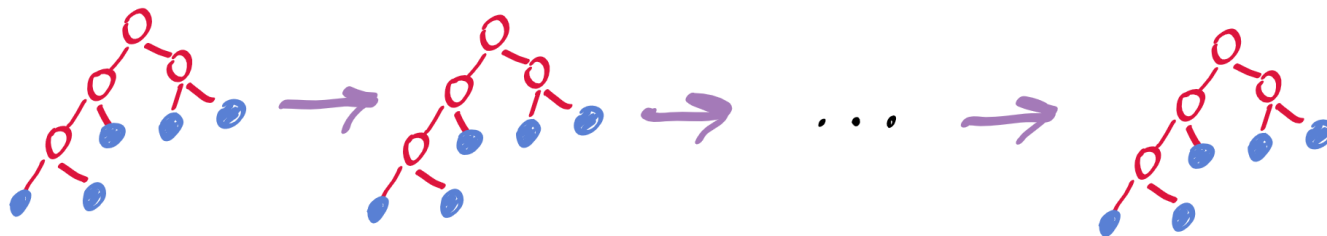
Classify with **majority voting**

RANDOM FOREST



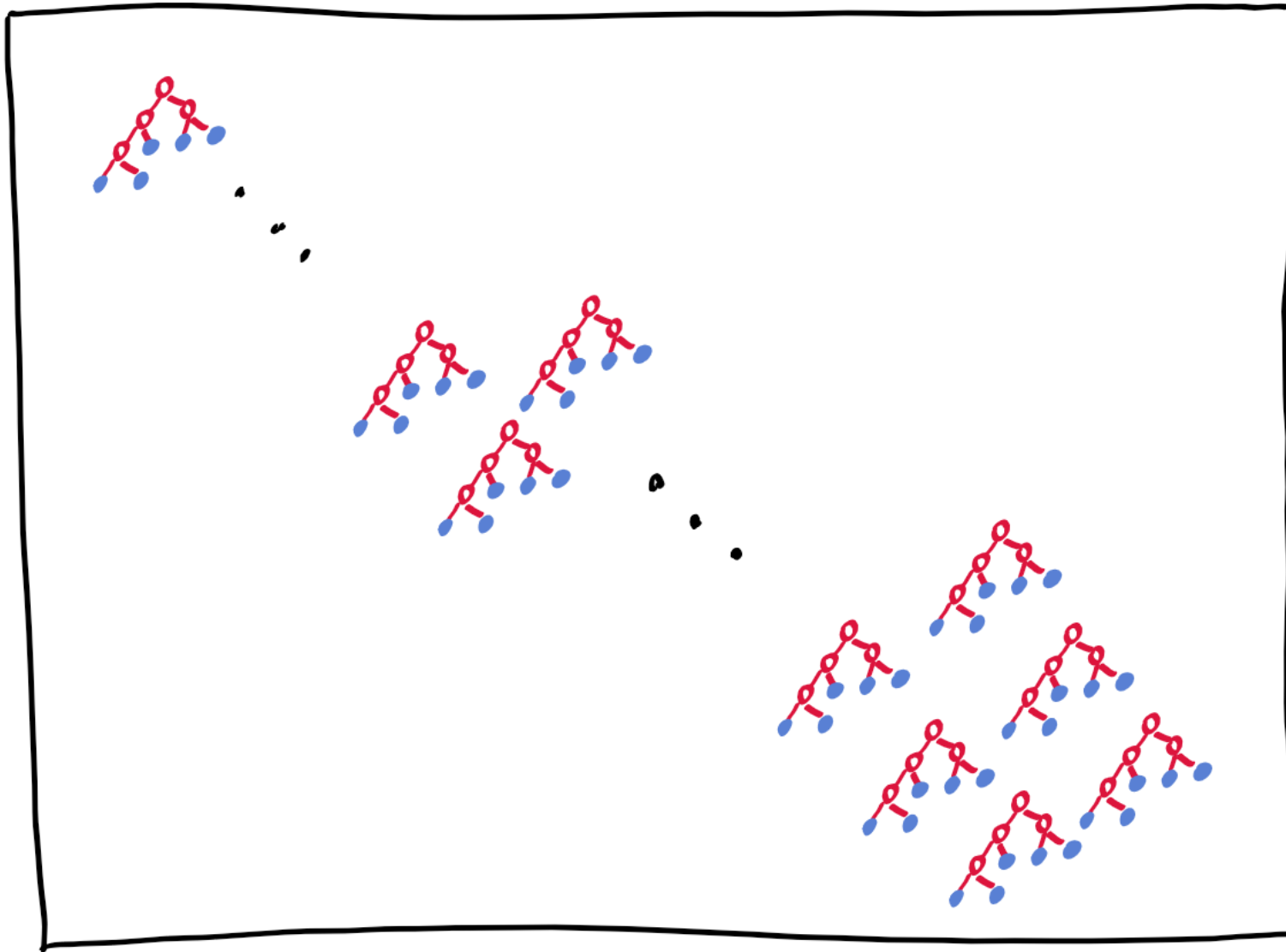
Classify with **majority voting**

BOOSTING

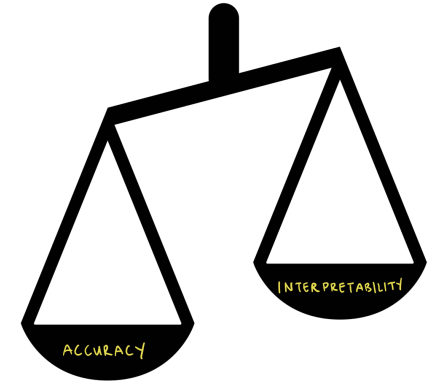


Classify with **classifier weights**
(tree)

Interpretability

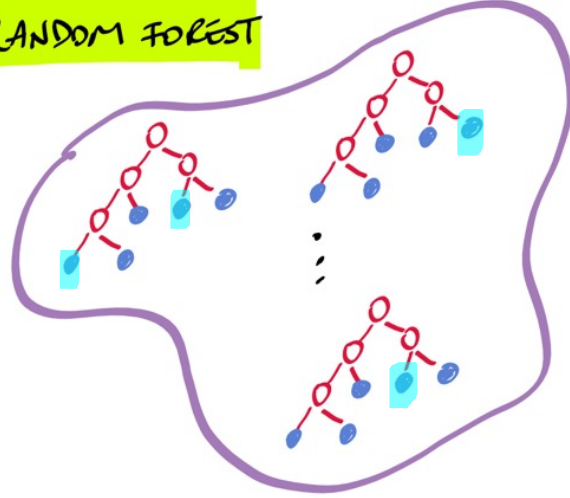


Accuracy

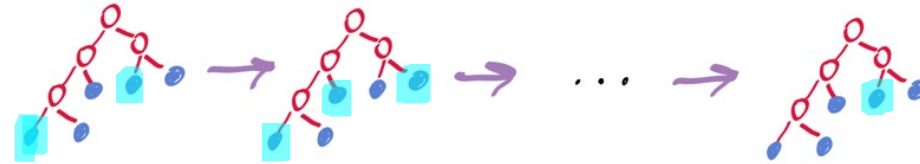


How to
Interpret?

RANDOM FOREST



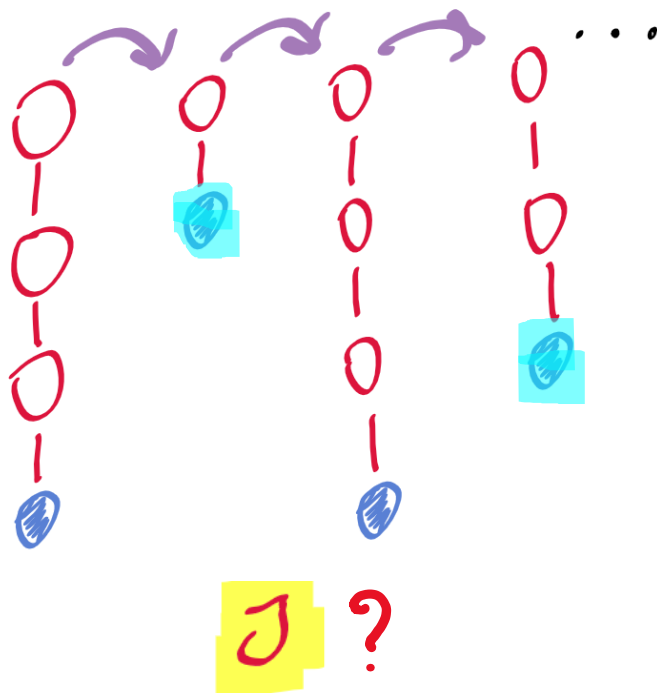
BOOSTING



J ?

RULE EXTRACTION

RUX



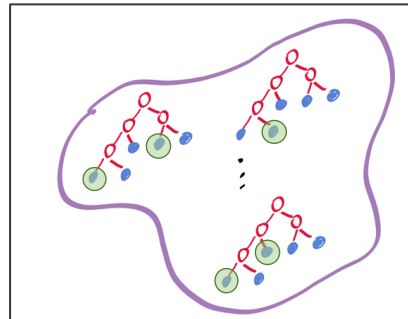
RULE GENERATION

RUG

$$\begin{array}{ll}
\text{minimize} & \lambda \sum_{j \in \mathcal{J}} c_j w_j + \sum_{i \in \mathcal{I}} v_i \\
\text{subject to} & \sum_{j \in \mathcal{J}} \hat{a}_{ij} w_j + v_i \geq 1, \quad i \in \mathcal{I}, \\
& v_i \geq 0, \quad i \in \mathcal{I}, \\
& w_j \geq 0, \quad j \in \mathcal{J},
\end{array}$$

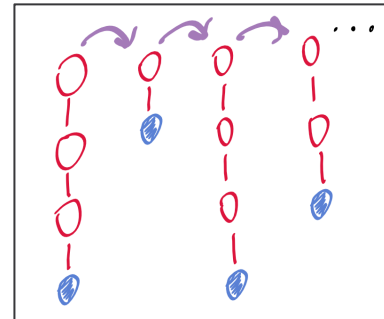
$$\begin{array}{ll}
\text{minimize} & \lambda \sum_{j \in \mathcal{J}} c_j w_j + \sum_{i \in \mathcal{I}} v_i \\
\text{subject to} & \sum_{j \in \mathcal{J}} \bar{a}_{ij} w_j + v_i \geq y_i, \quad i \in \mathcal{I}, \\
\text{subject to} & -\sum_{j \in \mathcal{J}} \bar{a}_{ij} w_j + v_i \geq -y_i, \quad i \in \mathcal{I}, \\
& v_i \geq 0, \quad i \in \mathcal{I}, \\
& w_j \geq 0, \quad j \in \mathcal{J}
\end{array}$$

Classification



Rule Extraction

Regression

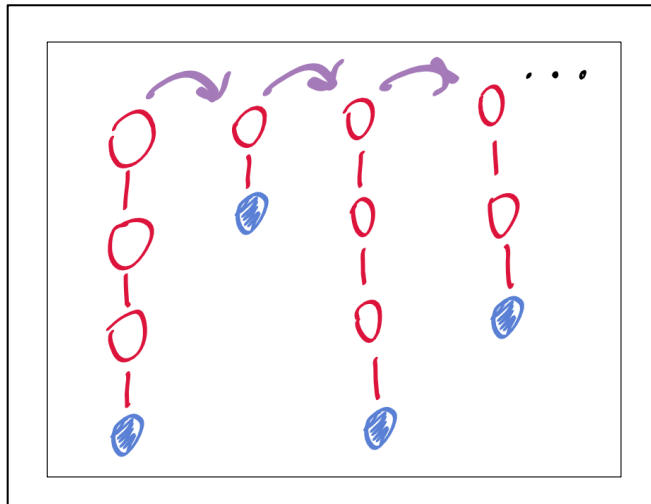


Rule Generation

RUG in Action

Loan Prediction: ~400K samples, 42 features

Rule Generation



Accuracy Scores

	<u>Acc.</u>	<u>F1</u>
Logistic Regression:	96%	0.79
Decision Tree:	95%	0.81
Random Forest:	97%	0.85
LightGBM:	97%	0.88
Rule Generation:	97%	0.88

Number of Rules

	<u>Rules</u>	<u>Depth</u>
Decision Tree:	4	2
Random Forest:	10791	7
LightGBM:	120	2
Rule Generation:	6	2
(Ave. num. of rules per prediction: 1.3)		

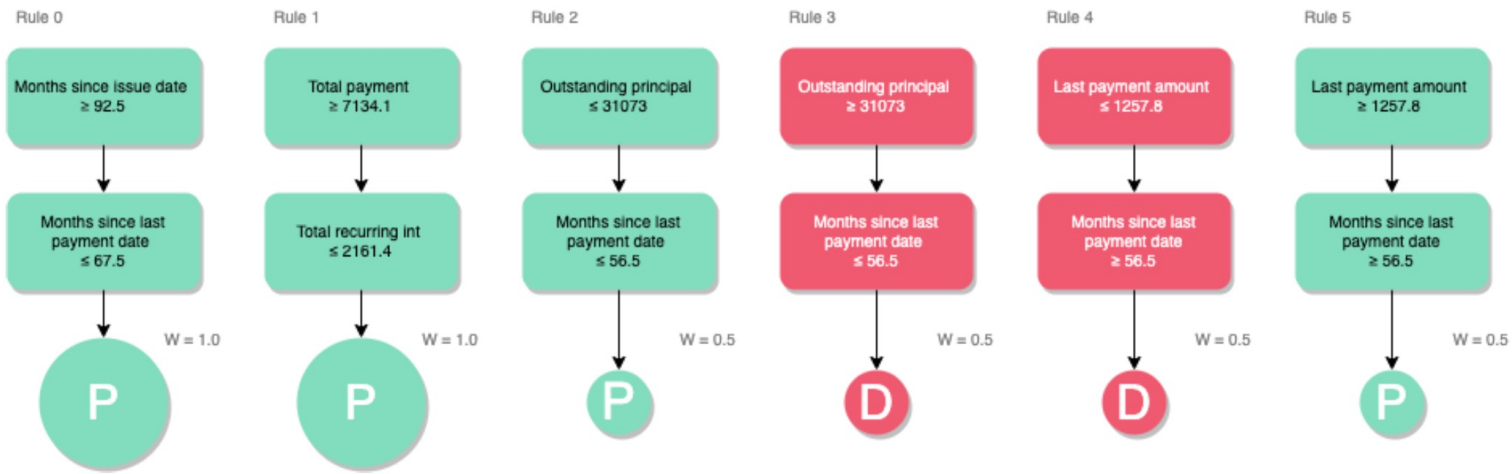
XAI models for Loan Prediction - Model Overview

Model scores 6 rules used over 11 of 42 features	
Accuracy	0.9759
F1 score	0.8804
Precision score	0.8811
Recall score	0.8797

Confusion matrix		
	Predicted D	Predicted P
Actual - D	70148	942
Actual - P	955	6985

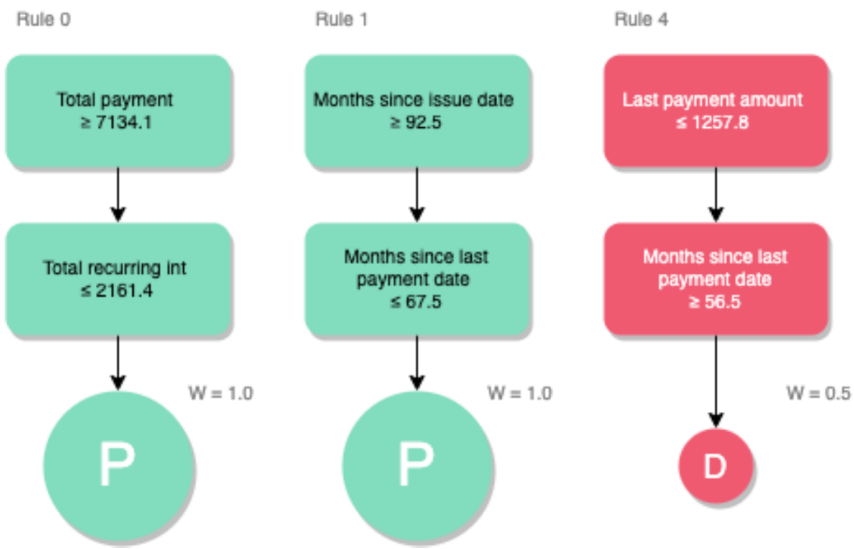
P - Performing D - Default

Model View



Prediction Features & Evaluation		Rule 0 Rule 1 Rule 4		
3 rules applied over 5 of 42 features				
Total payment	8436.7	P		
Total recurring int	1936.7	P		
Months since issue date	62		P	
Last payment amount	1936.7			D
Months since last payment date	62	P		D
Outstanding principal	0.0			
Annual income	46.000			
Total current balance	113.000			
Employment length	11			
Term	36			
Home ownership	yes			
Purpose - small business	yes			
Purpose - moving	no			
Purpose - dept consolidation	yes			
Purpose - home improvement	no			
* Please hover over class symbols to see their explanations				

Rule Breakdown



Final Prediction



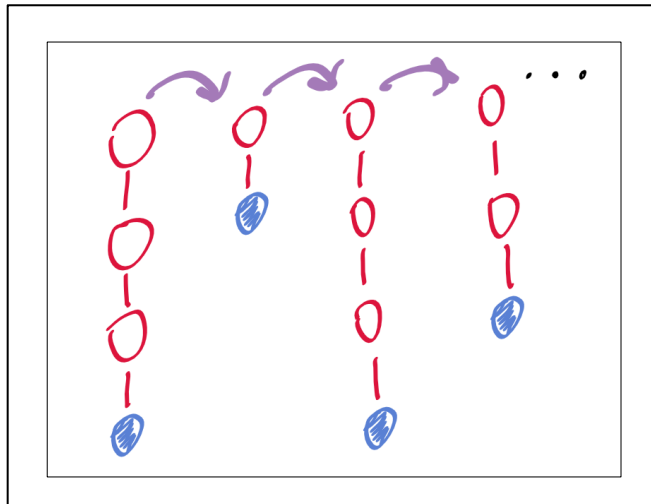
Performing is the final prediction of this specific observation (Customer id: **182977**). According to the majority voting among the results of the 3 applied rules, it received **80%** of the total vote (**2 out of 2.5**) and is chosen as the final prediction. This explanation is generated on 2022-06-07 18:03:05

P - Performing D - Default

RUG in Action

Loan Prediction: ~400K samples, 42 features

Rule Generation



If we prefer more accuracy!

Accuracy Scores

	<u>Acc.</u>	<u>F1</u>
Logistic Regression:	96%	0.79
Decision Tree:	99%	0.95
Random Forest:	99%	0.96
LightGBM:	99%	0.96
Rule Generation:	99%	0.95

Number of Rules

	<u>Rules</u>	<u>Depth</u>
Decision Tree:	1108	15
Random Forest:	255643	15
LightGBM:	1200	2
Rule Generation:	29	2
(Ave. num. of rules per prediction: 3.6)		